Mock API Creation

course-management-system-main/

│

├── frontend/

│

├── react-frontend/

│

└── mock-api/

└── db.json

db.json:


```
{
  "courses": [
    {
      "id": "1",
      "courseName": "Python Programming",
      "courseCode": "CS101",
      "instructor": "Dr. Anu",
      "duration": "8 Weeks",
      "level": "Beginner",
      "category": "Programming",
      "image": "https://cdn-icons-png.flaticon.com/512/5968/5968350.png",
      "status": "Active",
      "overview": "Learn Python programming from beginner to advanced level.",
      "learningOutcomes": [
        "Understand Python basics",
        "Work with functions",
```

```
    "Use object-oriented programming",
    "Build Python applications"
   ],
   "modules": [
    "Python Basics",
    "Control Statements",
    "Functions",
    "OOP",
    "Mini Project"
   ]
  },
  {
   "id": "2",
   "courseName": "Web Development",
   "courseCode": "WD101",
   "instructor": "Dr. Priya",
   "duration": "10 Weeks",
   "level": "Intermediate",
   "category": "Web Development",
   "image": "https://cdn-icons-png.flaticon.com/512/1006/1006363.png",
   "status": "Active",
   "overview": "Learn HTML, CSS and JavaScript for modern web development.",
   "learningOutcomes": [
    "Create semantic HTML",
    "Design responsive websites",
    "Use JavaScript",
    "Work with APIs"
   ],
```

```
  "modules": [
    "HTML",
    "CSS",
    "JavaScript",
    "DOM",
    "REST APIs"
  ]
 },
 {
  "id": "3",
  "courseName": "Artificial Intelligence",
  "courseCode": "AI101",
  "instructor": "Dr. Priya",
  "duration": "12 Weeks",
  "level": "Advanced",
  "category": "Artificial Intelligence",
  "image": "https://cdn-icons-png.flaticon.com/512/4712/4712027.png",
  "status": "Active",
  "overview": "Learn the fundamentals of Artificial Intelligence.",
  "learningOutcomes": [
    "Understand AI concepts",
    "Explore machine learning",
    "Understand neural networks",
    "Build AI applications"
  ],
  "modules": [
    "Introduction to AI",
    "Machine Learning",
```

```
      "Neural Networks",
      "Generative AI",
      "AI Project"
    ]
  }
 ]
}
```

Create API Service:

```
src/
└── services/
   └── api.js
```

```
import axios from "axios";
const api = axios.create({
  baseURL: "http://localhost:5000"
});
export default api;
```

Create Course Context:

```
src/
└── context/
   └── CourseContext.jsx
```

```
import {
  createContext,
  useContext,
  useEffect,
  useState
} from "react";
import api from "../services/api";
const CourseContext = createContext(null);
export function CourseProvider({ children }) {
  const [courses, setCourses] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);
  async function fetchCourses() {
    try {
      setLoading(true);
      const response =
        await api.get("/courses");
      setCourses(response.data);
      setError(null);
    } catch (error) {
      console.error(
        "Failed to fetch courses:",
        error
      );
      setError(
        "Unable to load courses."
      );
    } finally {
```

```
      setLoading(false);
    }
  }
  useEffect(() => {
    fetchCourses();
  }, []);
  async function addCourse(course) {
    const response = await api.post("/courses", course);
    setCourses((previousCourses) => [
      ...previousCourses,
      response.data
    ]);
  }
  async function updateCourse(
    id,
    updatedCourse
  ) {
    const response =
      await api.put(
        `/courses/${id}`,
        updatedCourse
      );
    setCourses((previousCourses) =>
      previousCourses.map(
        (course) =>
          course.id === id
            ? response.data
            : course
```

```
      )
    );
  }
  async function deleteCourse(id) {
    await api.delete(
      `/courses/${id}`
    );
    setCourses((previousCourses) =>
      previousCourses.filter(
        (course) =>
          course.id !== id
      )
    );
  }
  return (
    <CourseContext.Provider
      value={{
        courses,
        loading,
        error,
        fetchCourses,
        addCourse,
        updateCourse,
        deleteCourse
      }}
    >
      {children}
    </CourseContext.Provider>
```

```
  );
}
export function useCourses() {
  return useContext(
    CourseContext
  );
}
```

Modify main.jsx: ->Add course provider

```
import {
  createRoot
} from "react-dom/client";
import {
  BrowserRouter
} from "react-router-dom";
import App from "./App";
import {
  AuthProvider
} from "./auth/AuthContext";
import {
  CourseProvider
} from "./context/CourseContext";
createRoot(
  document.getElementById("root")
).render(
  <BrowserRouter>
```

```
      <AuthProvider>
        <CourseProvider>
          <App />
        </CourseProvider>
      </AuthProvider>
    </BrowserRouter>
);
```

Modify courses.jsx:

Currently having:

```
<CourseCard
  title="Python Programming"
  ...
/>
<CourseCard
  title="Web Development"
  ...
/>
<CourseCard
  title="Java Programming"
  ...
/>
```

Replace with:

```
import {
  useCourses
} from "../context/CourseContext";
```

Inside of the follwing code:

```
function Courses() {
  const {
    courses,
    loading,
    error
  } = useCourses();
```

Paste it:

```
{loading && (
  <p>Loading courses...</p>
)}
{error && (
  <p>{error}</p>
)}
{!loading &&
  !error &&
  courses.map((course) => (
    <CourseCard
      key={course.id}
      image={course.image}
      alt={course.courseName}
      title={course.courseName}
      description={course.overview}
      courseKey={course.id}
    />
  ))
}
```